6 Two people reaching at the same time for bread plates on opposite sides of them results in a pile-up that physicists would call a topological defect. In this specific case, it would be a domain wall, which, if let loose on the cosmos, would dominate the universe and lead to a Big Crunch. This is why I always wait for someone else to select the bread before making my attempt.

7 We previously discussed this transition, and what it meant for the very early universe, in Chapter 2.

8 Of course, the Higgs field doesn't have preferences; it's just governed by its potential. But the way it would dive into a true vacuum if it could would definitely give an impression of enthusiasm.

9 "Some of us" here being, specifically, me and my colleague Robert McNees, in our 2018 paper in *Physical Review D*. It was a fun one.

10 This discussion remains, to me, one of the most beautiful pieces of physics poetry I've ever seen in an academic journal.

## 7. Bounce

1 "Simple" here may be a matter of perspective. Working with the equations of general relativity requires a deep understanding of differential geometry, which is the sort of thing you pretty much only study if you're doing graduate work in physics or mathematics. But if you ARE such a person, the equations are as elegant and transparent as fine-blown glass.

2 That thing where a first-draft theory is dramatically redesigned but still useful applies to inflation too. The original version of inflation is widely considered to be a stroke of genius in spite of being, in the end, a total failure. It didn't work at all, and was totally revamped by other physicists within about a year. What its originators did exactly right was to propose a general class of solutions that became the spark of a firestorm of creative ways to finally make the Big Bang work. The revamped version, what we